

UMBRELA-Indo-IR — Catatan Eksperimen Lengkap

Tugas Kelompok IR Genap 2025/2026

Tim: Faiz · Arvin · Karolina (Karol) · Vincent (Radit) · Radit (Arya/Raditya)

Dokumen ini merangkum **semua eksperimen** yang telah dilakukan, mulai dari qrel generation, evaluasi retrieval, training reranker, fine-tuning LLM judge, hingga extended analysis. Ditulis sebagai catatan teknis yang cukup rinci untuk dapat mereproduksi semua hasil.

Daftar Isi

- 1. Dataset & Setup
- 2. RQ1 — LLM Judge Agreement
- 3. RQ2 — Retrieval & Reranking
- 4. RQ3 — Bias Analysis
- 5. Extended Analysis
- 6. Kontribusi Per Anggota
- 7. Lokasi File & Artifacts

1. Dataset & Setup

1.1 MIRACL-ID

Item	Detail
Dataset	MIRACL (Multilingual Information Retrieval Across a Continuum of Languages) — split Indonesia
Corpus	~1.44 juta passage dari Wikipedia Bahasa Indonesia
Format corpus	JSONL, field: <code>docid</code> , <code>doc</code>
Test split	960 queries dengan human qrels (binary, 0/1, positive-only format)
Train split	3,257 queries (dipakai untuk hard negative mining & reranker training)
Val split	814 queries (dipakai untuk hyperparameter tuning)
Human qrels	3,088 pairs total di test split, semua score=1 (positive-only)
Lokasi	<code>data/miracl-id/</code>

Penting tentang format human qrels: MIRACL human qrels hanya menyimpan dokumen yang dinilai relevan (positive-only). Dokumen yang tidak muncul di qrels dianggap tidak relevan secara implisit. Ini berbeda dari LLM qrels yang menyimpan semua 9,668 pasang query-dokumen beserta skor 0–3.

1.2 Lingkungan & Tools

Tool	Versi / Keterangan
Python	3.12
sentence-transformers	5.5.0
ranx	evaluasi IR (nDCG@10, MAP@10, Recall@100)

Tool	Versi / Keterangan
bm25s	BM25 retrieval tanpa Java
faiss	Dense retrieval (FAISS IVF index)
vLLM	Inference LLM (Llama3, Gemma2)
TRL	SFTTrainer untuk LoRA fine-tuning
scikit-learn	Cohen's κ
Platform training	vast.ai (GPU A100/H100)

1.3 Prompt UMBRELA

Proyek ini menggunakan prompt UMBRELA yang diadaptasi ke Bahasa Indonesia. Ada beberapa varian prompt yang diuji (lihat RQ1 ablation):

- **zeroshot_bing** (default, terbaik): Prompt zero-shot dengan gaya Bing, meminta skor 0–3
- **zeroshot_bing_strict**: Versi ketat (kriteria relevansi lebih strict)
- **fewshot_bing**: Versi few-shot dengan contoh
- **zeroshot_basic**: Prompt minimal tanpa konteks web-search
- **fewshot_basic**: Few-shot versi minimal

2. RQ1 — LLM Judge Agreement

Pertanyaan: LLM judge mana yang paling konsisten dengan penilaian human assessor untuk IR berbahasa Indonesia?

Metrik: Cohen's κ (inter-rater agreement, chance-corrected), positif rate LLM vs human, jumlah pairs

2.1 Qrel Generation (Faiz — Qwen2.5-7B)

Model: Qwen/Qwen2.5-7B-Instruct

Proses:

1. Ambil top-100 kandidat dari BM25 untuk setiap query (9,668 pairs untuk test split)
2. Format setiap pasang (query, passage) ke dalam prompt UMBRELA
3. Jalankan inference dengan vLLM, ekstrak skor 0–3
4. Simpan sebagai TREC qrels format: `qid 0 docid score`

Hasil per split:

Split	n_pairs	κ vs Human	LLM pos rate	Human pos rate
Train	33,076	0.3039	30.02%	30.17%
Val	8,282	0.2860	29.11%	30.50%
Test	9,668	0.3767	30.74%	31.94%

Threshold: $\tau=2$ (binary: score $\geq 2 \rightarrow$ relevan). Ini dikonfirmasi optimal via calibration sweep (lihat §2.6).

File:

- `data/miracl-id/results/qrels/qwen_train.txt, qwen_val.txt, qwen_test.txt`

- `results/final/kappa_qwen_train.csv`, `kappa_qwen_val.csv`, `kappa_qwen_test.csv`
-

2.2 Qrel Generation (Vincent/Radit — SahabatAI-Gemma2)

Model: `SahabatAI/sahabatai-v1-gemma2-9b-instruct` (Gemma2-9B, Indonesian-tuned)

Proses: Inference menggunakan vLLM di vast.ai (GPU A100).

Hasil per split:

Split	n_pairs	κ vs Human	LLM pos rate	Human pos rate
Train	33,076	0.3222	40.77%	30.17%
Val	8,282	0.3204	39.37%	30.50%
Test	9,668	0.3763	41.23%	31.94%

Catatan: Gemma2 sedikit overpredicts (pos rate 41% vs human 32%) tapi κ kompetitif dengan Qwen.

File:

- `results/qrels/sahabat_llama_test.txt` (nama file historical, isinya Gemma2)
 - `results/final/kappa_gemma.csv`
-

2.3 Qrel Generation (Vincent — SahabatAI-Llama3)

Model: `SahabatAI/sahabatai-v1-llama3-8b-instruct` (Llama3-8B, Indonesian-tuned)

Proses: Inference unquantized di vast.ai.

Hasil:

Prompt Mode	n_pairs	κ vs Human	LLM pos rate
default (zeroshot_bing)	9,668	0.2103	66.66%
strict	9,668	0.3652	38.79%

Temuan kritis: Tanpa prompt strict, Llama3 menilai 67% dokumen sebagai relevan → κ sangat rendah (0.21). Dengan prompt strict yang membatasi output format, κ naik ke 0.37 dan pos rate turun ke 39% (lebih mendekati human).

File:

- `results/qrels/sahabat_llama_test.txt` (default)
 - `results/qrels_strict/sahabat_llama_strict_test.txt` (strict)
 - `results/final/kappa_llama.csv`, `kappa_llama_strict.csv`
-

2.4 Qrel Generation — DeepSeek-V3 & ChatGPT (Arvin)

Model DeepSeek: `deepseek-ai/DeepSeek-V3` (API)

Model ChatGPT: `gpt-4o-mini` (API)

Hasil:

Judge	n_pairs	κ vs Human	LLM pos rate
DeepSeek-V3	9,668	0.4219	27.99%
ChatGPT (gpt-4o-mini)	8,750	0.3856*	26.38%

*ChatGPT hanya 8,750 dari 9,668 pairs (ada ~920 query tanpa judgement karena API error/limit)

DeepSeek-V3 = judge terbaik ($\kappa=0.42$), paling konservatif (28% pos rate), paling mendekati human (32%).

File:

- [results/qrels/deepseek_test.txt](#)
- [results/qrels/chatgpt_test.txt](#)
- [results/final/kappa_deepseek_test.csv](#)

2.5 Prompt Ablation (Faiz — Qwen2.5-7B, 5 Variant)

Pertanyaan: Apakah format prompt berpengaruh signifikan terhadap κ ?

Setup: Semua 5 varian dijalankan di test split (9,668 pairs) dengan model Qwen2.5-7B-Instruct.

Prompt Mode	κ	LLM pos rate
zeroshot_bing ← baseline terbaik	0.3767	30.74%
zeroshot_bing_strict	0.3218	23.13%
fewshot_bing	0.2720	16.39%
zeroshot_basic	0.2393	12.65%
fewshot_basic	0.2254	11.46%

Kesimpulan: **zeroshot_bing** terbaik. Few-shot justru menurunkan κ secara signifikan — kemungkinan karena contoh few-shot membuat model terlalu selektif (pos rate 11–16%).

File: [results/final/kappa_prompt_ablation.csv](#)

2.6 Prompt Ablation — Gemma2 & Llama3 (Vincent, extended)

Ablation dilanjutkan untuk SahabatAI-Gemma2 dan SahabatAI-Llama3 menggunakan vLLM.

SahabatAI-Gemma2 (vLLM):

Prompt Mode	κ	LLM pos rate
zeroshot_bing	0.3778	41.03%
zeroshot_bing_strict	0.3753	45.29%
zeroshot_basic	0.3628	45.36%
fewshot_bing	0.3843	41.62%
fewshot_basic	0.3121	53.57%
zeroshot_basic (partial, 2,151 pairs)	0.3491	29.66%

SahabatAI-Llama3 (vLLM):

Prompt Mode	κ	LLM pos rate
fewshot_bing	0.3415	34.90%
fewshot_basic	0.3328	44.89%
zeroshot_basic	0.3039	58.06%

File: [results/final/kappa_prompt_ablation_full.csv](#)

2.7 Calibration Analysis — Threshold Sweep (Faiz)

Pertanyaan: Apakah $\tau=2$ sudah optimal untuk binarisasi skor LLM?

Setup: Sweep $\tau \in \{1, 2, 3\}$ pada val dan test split, Qwen2.5-7B.

τ	κ (val)	κ (test)	LLM pos rate (test)	Optimal?
1	0.2541	0.3255	52.14%	×
2	0.2860	0.3767	30.74%	✓
3	0.1459	0.1808	7.60%	×

Kesimpulan: $\tau=2$ sudah optimal. Masalah κ bukan di thresholding tapi di per-pair reasoning.

File: [results/final/calibration_qwen.csv](#) Script: [evaluation/calibrate.py](#)

2.8 LoRA Fine-tuning Qwen2.5-7B sebagai Judge (Faiz + Arvin)

Pertanyaan: Apakah fine-tuning Qwen2.5-7B dengan human qrels sebagai label meningkatkan κ ?

Data:

- SFT training data: 33,076 pairs dari train split
- Format: **prompt** (query + passage formatted) → **response** (skor relevansi)
- Total: ~33k train examples + 8k val examples
- Lokasi data: [results/lora_data/qwen/](#)

Training:

- Base model: [Qwen/Qwen2.5-7B-Instruct](#)
- Method: LoRA via TRL [SFTTrainer](#)
- Adapter: [lora/](#) → [results/models/lora_qwen_smoke/](#)
- Epochs: 3, bfloat16
- HF push: [fassabilf/lora-qwen-miracle-id-smoke](#) (private)

Hasil:

System	κ (test)	LLM pos rate
Qwen baseline (zeroshot_bing)	0.3767	30.74%
Qwen LoRA fine-tuned	0.3718	14.96%
DeepSeek-V3 (best judge overall)	0.4219	27.99%

Temuan: LoRA fine-tuning tidak meningkatkan κ secara signifikan (0.3718 vs 0.3767, perbedaan kecil dalam batas noise). Model LoRA menjadi jauh lebih konservatif (pos rate turun dari 30.7% ke 15%), tetapi kappa tidak meningkat. Ini mengindikasikan bahwa sinyal biner dari human qrels tidak cukup kaya untuk mendorong improvement via LoRA; model butuh sinyal yang lebih nuanced.

ORPO fine-tuning: Script sudah disiapkan ([lora/train_orpo.py](#)), ORPO data prep sudah ada di [results/orpo_data/qwen/](#). Experiment belum selesai (belum ada hasil final).

File: [results/final/kappa_qwen_lora_test.csv](#) **Script:** [lora/train.py](#), [lora/prepare_data.py](#), [lora/train_orpo.py](#), [lora/prepare_orpo_data.py](#)

2.9 Ringkasan RQ1 — Semua Judge

Judge	Model	κ (test)	LLM pos rate	n_pairs
DeepSeek-V3	deepseek-ai/DeepSeek-V3	0.4219	27.99%	9,668
ChatGPT (gpt-4o-mini)	gpt-4o-mini	0.3856	26.38%	8,750
SahabatAI-Gemma2 (zeroshot_bing)	sahabatai-v1-gemma2-9b	0.3778	41.03%	9,668
Gemma2 (zeroshot_bing_strict)	sahabatai-v1-gemma2-9b	0.3753	45.29%	9,668
Qwen2.5-7B (zeroshot_bing)	Qwen2.5-7B-Instruct	0.3767	30.74%	9,668
Qwen2.5-7B LoRA fine-tuned	Qwen2.5-7B + LoRA	0.3718	14.96%	9,668
Gemma2 (fewshot_bing)	sahabatai-v1-gemma2-9b	0.3843	41.62%	9,668
Llama3-strict	sahabatai-v1-llama3-8b	0.3652	38.79%	9,668
Gemma2 (zeroshot_basic)	sahabatai-v1-gemma2-9b	0.3628	45.36%	9,668
Llama3-vllm-fewshot-bing	sahabatai-v1-llama3-8b	0.3415	34.90%	9,668
Llama3-vllm-fewshot-basic	sahabatai-v1-llama3-8b	0.3328	44.89%	9,668
Llama3-vllm-zeroshot-basic	sahabatai-v1-llama3-8b	0.3039	58.06%	9,668
Llama3-default	sahabatai-v1-llama3-8b	0.2103	66.66%	9,668

Temuan utama RQ1:

- DeepSeek-V3 adalah judge terbaik ($\kappa=0.42$), juga paling konservatif
- Qwen2.5-7B terbaik di antara open-source yang bisa dijalankan sendiri ($\kappa=0.38$)
- Llama3 wajib menggunakan prompt strict; tanpa itu κ hanya 0.21
- LoRA fine-tuning tidak memberikan improvement signifikan

3. RQ2 — Retrieval & Reranking

Pertanyaan: Bisakah qrels yang dihasilkan LLM melatih reranker yang meningkatkan BM25?

3.1 First-Stage Retrieval (Arvin)

Sistem yang diimplementasikan:

Sistem	Library	Indexing	Keterangan
BM25	bm25s	—	Tanpa Java, pure Python

Sistem	Library	Indexing	Keterangan
BGE-M3 dense	BAAI/bge-m3	FAISS IVF	XLM-RoBERTa multi-vector
Hybrid BM25+BGE-M3	bm25s + FAISS	RRF	Reciprocal Rank Fusion
Qwen-embed	Qwen embedding (old)	FAISS	Legacy, rendah overlap human
Qwen3-embed	Qwen/Qwen3-Embedding-4B	FAISS	Tanpa instruction prefix (Karol)
Hybrid BM25+Qwen3	bm25s + FAISS	RRF	Qwen3 + BM25 fusion (Karol)

Hasil retrieval (nDCG@10, human qrels, test split, 960 queries):

Retriever	nDCG@10	MAP@10	Recall@100	Queries nDCG=0
BGE-M3	0.5604	0.4529	0.9047	116/960 (12.1%)
Hybrid BM25+BGE-M3	0.5191	—	0.9154	—
Qwen3-embed (no instr)	0.4958	0.3913	0.8508	156/960 (16.2%)
Hybrid BM25+Qwen3	0.4603	0.3502	0.8792	161/960 (16.8%)
BM25	0.3053	0.2206	0.7634	335/960 (34.9%)
Qwen-embed (legacy)	~0.007	~0.004	~0.04	—

Catatan Qwen-embed legacy: Nilai sangat rendah karena pool bias — kandidat Qwen-embed hampir tidak ada yang overlap dengan dokumen yang di-judge oleh human (pooling MIRACL dilakukan dari BM25/BGE pool, bukan Qwen-embed).

Catatan Qwen3-embed: Dievaluasi tanpa instruction prefix (lower bound). Dengan instruction prefix, nDCG@10 di val split naik +0.059 (dari 0.4336 → 0.4923) — jadi hasil test juga diperkirakan lebih tinggi ~0.05.

File candidates:

- `candidates/bm25_test_top100.jsonl`
- `candidates/bgem3_test_top100.jsonl`
- `candidates/hybrid_test_top100.jsonl` (BM25+BGE)
- `candidates/qwen_test_top100.jsonl` (Qwen-embed legacy)
- `candidates/qwen3_test_top100.jsonl` (Qwen3-embed, no instruction)
- `candidates/hybrid_bm25_qwen3_test_top100.jsonl`

nDCG@K breakdown (4 retriever utama):

Retriever	@1	@3	@5	@10
BGE-M3	0.5563	0.5168	0.5189	0.5604
Qwen3-embed	0.4844	0.4537	0.4647	0.4958
Hybrid BM25+Qwen3	0.3979	0.3914	0.4156	0.4603
BM25	0.2437	0.2470	0.2618	0.3055

3.2 Qwen3-embed Instruction Prefix Ablation (Karol)

Pertanyaan: Apakah menambah instruction prefix ke Qwen3-embed signifikan?

Val split, 814 queries:

Setup	nDCG@10	Recall@10	Recall@100	MRR
no-instruction	0.4336	0.4604	0.7910	0.5498
with-instruction	0.4923	0.5238	0.8614	0.6095
BM25 baseline	0.2938	0.3290	0.7257	0.3658

Delta: +0.059 nDCG@10 | +0.070 Recall@100 | +0.060 MRR

Ini sangat signifikan — instruction prefix memberikan improvement hampir seperti selisih antara Qwen3 dan Hybrid. Semua evaluasi test split Qwen3-embed adalah **lower bound**.

3.3 Reranker Training dari LLM Qrels — Size Ablation (Radit/Vincent)

Pertanyaan: Berapa banyak training data dari LLM qrels yang dibutuhkan untuk melatih reranker efektif?

Setup:

- First-stage: BM25 (baseline 0.3053)
- Base reranker model: `BAAI/bge-reranker-v2-m3` (XLM-RoBERTa for sequence classification)
- Training data: qrels SahabatAI-Gemma2 (format: triplet positif/negatif dari top-100 BM25)
- Loss: Binary Cross-Entropy
- Epochs: 3, batch 16, lr 5e-5

Hasil size ablation (test split, BM25 first-stage):

N queries	N triplets	N train examples	nDCG@10	MAP@10	Recall@100	Val AP (LLM)
100	1,937	3,874	0.5178	0.4088	0.7634	0.865
300	5,285	10,570	0.4620	0.3491	0.7634	0.873
500	9,173	18,346	0.5011	0.3950	0.7634	0.905
1000	18,509	37,018	0.4072	0.2993	0.7634	0.916
full (3,257)	60,750	121,500	0.3993	0.2917	0.7634	1.000

Temuan utama:

- **N=100 menghasilkan nDCG@10 tertinggi (0.5178)** — lebih baik dari N=full (0.3993)
- Val AP pada LLM qrels naik monoton ke 100% → reranker **overfitting** pada noise LLM judge
- BM25 + Gemma2-rk (N=100): 0.5178 ≈ BGE-M3 baseline 0.5604 (selisih 0.04)
- **Counterintuitive:** Lebih banyak data LLM = lebih rendah nDCG@10 vs human qrels

Model: Semua 5 model tersimpan di `results/models/reranker_{100,300,500,1000,full}/`
File: `results/final/ablation_summary.csv, size_{100,300,500,1000,full}.json`

3.4 Reranker Training dari LLM Qrels — Qwen (Faiz)

Setup:

- First-stage: BM25
- Base model: `BAAI/bge-reranker-v2-m3`

- Training data: Qwen2.5-7B qrels (full dataset: 53,727 triplets)
- Epochs: 3, bf16, max-length 256, lr 2e-5

Hasil:

System	nDCG@10	MAP@10	Recall@100
BM25 baseline	0.3053	—	0.7634
BM25 + Qwen-rk	0.4478	—	0.7634

Delta vs BM25: +0.1425 (+46.7%)

Model: [results/models/reranker_qwen/](#) → HF: [fassabilf/umbrela-indo-ir-reranker-qwen](#) (public)

3.5 Hard Negative Mining (Karol)

Pertanyaan: Apakah reranker yang dilatih dari hard negatives (bukan LLM qrels) lebih efektif?

Konsep hard negative mining:

- Ambil top-20 kandidat retriever
- Dokumen yang ada di top-20 tapi dinilai tidak relevan oleh human (atau tidak muncul di human qrels) = hard negative
- Ini lebih informatif dari random negative karena secara semantik mirip query

Hard negatives dari Qwen3-embed:

- Queries: 3,257 (semua train)
- Triplets: **53,727**
- Lokasi: [results/reranker_data/qwen/](#)
- Model dilatih: [results/models/reranker_qwen/](#) (dengan Qwen3 hard negatives)

Hard negatives dari BGE-M3:

- Queries: 3,257 (semua train)
- Explicit label=0 hard negs: 8,416
- Unannotated hard negs: 49,486
- Total triplets: **130,961** (2.4x lebih banyak dari Qwen3 hard negs)
- Lokasi: [results/reranker_data/bgem3_hardneg/](#)
- Model dilatih: [results/models/reranker_bgem3_hardneg/](#) → HF: [karolinajocelyn/umbrela-indo-ir-models](#) (BGE-hardneg-rk)

Mengapa BGE-M3 punya lebih banyak hard negs?

BGE-M3 punya Recall@100 lebih tinggi (0.9047 vs 0.8508) → lebih banyak dokumen masuk top-20 → lebih banyak false positives yang bisa dijadikan hard negative.

Script: [reranker/mine_hard_negatives.py](#)

3.6 Hasil Lengkap Reranking (semua kombinasi, Karol)

Evaluasi: nDCG@10 dengan human qrels, test split, 960 queries.

First-Stage	Reranker	Training Signal	nDCG@10	Δ abs	Δ %	Recall@100
BM25	—	—	0.3053	—	—	0.7634

First-Stage	Reranker	Training Signal	nDCG@10	Δ abs	Δ %	Recall@100
BM25	Qwen-rk	Qwen LLM qrels	0.4478	+0.1425	+46.7%	0.7634
BM25	Gemma2-rk (N=100)	Gemma2 LLM qrels	0.5178	+0.2125	+69.6%	0.7634
BGE-M3	—	—	0.5604	—	—	0.9047
BGE-M3	Qwen-rk	Qwen LLM qrels	0.4495	-0.1109	-19.8%	0.9047
BGE-M3	Gemma2-rk	Gemma2 LLM qrels	0.5111	-0.0493	-8.8%	0.9047
BGE-M3	Qwen3- hardneg-rk	Hard negs (Qwen3)	0.5276	-0.0328	-5.9%	0.9047
BGE-M3	BGE-hardneg- rk	Hard negs (BGE-M3)	0.5659	+0.0055	+1.0%	0.9047
Qwen3-embed	—	—	0.4958	—	—	0.8508
Qwen3-embed	Qwen-rk	Qwen LLM qrels	0.4585	-0.0373	-7.5%	0.8508
Qwen3-embed	Gemma2-rk	Gemma2 LLM qrels	0.5160	+0.0202	+4.1%	0.8508
Qwen3-embed	Qwen3- hardneg-rk	Hard negs (Qwen3)	—	—	—	—
Qwen3-embed	BGE-hardneg- rk	Hard negs (BGE-M3)	0.5882	+0.0924	+18.6%	0.8508
Hybrid BM25+Qwen3	—	—	0.4603	—	—	0.8792
Hybrid BM25+Qwen3	Qwen-rk	Qwen LLM qrels	0.4604	+0.0001	~0%	0.8792
Hybrid BM25+Qwen3	Qwen3- hardneg-rk	Hard negs (Qwen3)	0.5306	+0.0703	+15.3%	0.8792

Recall@100 tidak berubah setelah reranking — reranker hanya mengubah urutan, tidak menambah/mengurangi kandidat.

File JSON hasil eval:

- [results/final/bm25_qwen_rk.json](#) — BM25 + Qwen-rk
- [results/final/bgem3_qwenrk.json](#) — BGE-M3 + Qwen-rk
- [results/final/bgem3_gemma2rk_test.json](#) — BGE-M3 + Gemma2-rk
- [results/final/bgem3_qwen3hardnegrk_test.json](#) — BGE-M3 + Qwen3-hardneg-rk
- [results/final/bgem3_bgem3hardnegrk_test.json](#) — BGE-M3 + BGE-hardneg-rk ★
- [results/final/qwen3_qwenrk.json](#) — Qwen3 + Qwen-rk
- [results/final/qwen3_gemma2rk_test.json](#) — Qwen3 + Gemma2-rk
- [results/final/qwen3_bgem3hardnegrk_test.json](#) — Qwen3 + BGE-hardneg-rk ★
- [results/final/hybrid_bm25_qwen3_test.json](#) — Hybrid (no reranker)
- [results/final/hybrid_bm25_qwen3_qwenrk.json](#) — Hybrid + Qwen-rk
- [results/final/hybrid_qwen3_hardnegrk_test.json](#) — Hybrid + Qwen3-hardneg-rk

4. RQ3 — Bias Analysis

Pertanyaan: Apakah pilihan LLM judge/reranker dari family X memberikan keuntungan sistematis untuk retriever dari family yang sama? Apakah training signal lebih menentukan dari family alignment?

4.1 Reranker Bias — Temuan

Lihat §3.6 untuk semua angka. Temuan kunci:

1. Qwen-rk merusak BGE-M3 lebih parah dari Qwen3-embed (-19.8% vs -7.5%)

→ Ini adalah asimetri yang mengindikasikan bias, tapi arahnya paradoks: Qwen-rk (family Qwen) justru merusak Qwen3-embed (-7.5%) dan BGE-M3 (-19.8%) — tidak ada yang diuntungkan.

2. BGE-hardneg-rk membantu Qwen3 lebih besar (+18.6%) dari BGE-M3 sendiri (+1.0%)

→ Reranker yang dilatih dari kesalahan retriever kuat menghasilkan sinyal yang bersifat general/transfer.

3. Hard negative mining > LLM-judged qrels

→ Perbedaan antara BGE-hardneg-rk (+18.6%) dan Qwen-rk (-7.5%) jauh lebih besar dari variasi antar-family.

4.2 Hard Negative Overlap Analysis

Seberapa berbeda kandidat BGE-M3 vs Qwen3 pada query yang sama?

K	Overlap Rate	Excl BGE-M3 hard-negs	Excl Qwen3 hard-negs	% BGE-M3 HN tidak di Qwen3
10	49.2%	4.65/query	4.91/query	57.1%
20	47.0%	10.16/query	10.44/query	57.3%
50	43.4%	27.93/query	28.19/query	59.0%
100	40.5%	59.23/query	59.44/query	61.0%

Interpretasi: Di K=10, hanya 49.2% dokumen yang sama antara BGE-M3 dan Qwen3. Masing-masing retriever punya blind spot yang berbeda dan hampir seimbang jumlahnya (4.65 vs 4.91 per query di K=10).

Implikasi untuk reranker: Ketika reranker dilatih dari hard negs salah satu retriever, 57% dari apa yang dipelajarinya tidak relevan untuk retriever lain (dokumen tersebut tidak ada di kandidat retriever lain). Ini menjelaskan mengapa Qwen3-hardneg-rk kurang efektif pada BGE-M3. Sebaliknya, BGE-M3 hard negs masih ~41% overlap dengan kandidat Qwen3, cukup untuk memberikan transfer.

File: results/final/bias_analysis/hardneg_overlap_report.md

4.3 Judge Bias Analysis — Matrix nDCG@10 per Judge per Retriever

Semua nilai dihitung dengan threshold ≥ 2 kecuali Human (≥ 1)

Judge	BM25	BGE-M3	Qwen3-embed	Hybrid BM25+Qwen3	BGE-Qwen3 Δ
Human	0.3055	0.5604	0.4958	0.4603	-0.0647
DeepSeek-V3	0.2980	0.5805	0.5063	0.4642	-0.0739
Qwen2.5-7B	0.2833	0.5687	0.4988	0.4499	-0.0700
ChatGPT	0.2662	0.5525	0.4751	0.4246	-0.0774
Gemma2-vllm-zs-bing	0.2543	0.5006	0.4299	0.3981	-0.0707

Judge	BM25	BGE-M3	Qwen3-embed	Hybrid BM25+Qwen3	BGE-Qwen3 Δ
Gemma2-vllm-zs-bing-strict	0.2474	0.4713	0.4012	0.3803	-0.0701
Llama3-strict	0.2521	0.4757	0.4072	0.3877	-0.0685
Llama3-vllm-fs-basic	0.2506	0.4903	0.4245	0.3934	-0.0658
Llama3-vllm-fs-bing	0.2422	0.4949	0.4377	0.3930	-0.0572
Llama3-vllm-zs-basic	0.2606	0.4887	0.4164	0.3985	-0.0723
Llama3-default	0.2352	0.4241	0.3552	0.3542	-0.0689

Temuan utama judge bias:

- **Ranking konsisten di semua judge:** BGE-M3 > Qwen3 > Hybrid > BM25 — tidak ada judge yang "membalik" leaderboard
- **BGE-Qwen3 Δ selalu negatif** (range -0.06 s/d -0.08) — semua judge setuju BGE-M3 lebih baik
- **Tidak ada bukti same-family bias** — Qwen2.5-7B judge tidak menaikkan Qwen3-embed relatif terhadap human assessment
- **DeepSeek menginflasi skor absolut** (+9% untuk BGE-M3) tapi tidak mengubah ranking
- **ChatGPT deflate skor Hybrid** (0.4246 vs Human 0.4603) — paling skeptis terhadap hybrid retrieval
- **Llama3-default sangat rendah** — terpengaruh parah oleh pos rate 67%

File: [results/final/bias_analysis/judge_matrix.md](#), [results/final/extended/full_matrix.csv](#)

MD analisis lengkap: [results/final/rq3_bias_analysis.md](#)

5. Extended Analysis

Semua output di [results/final/extended/](#). Dijalankan via [evaluation/extended_analysis.py](#).

5.1 Inter-Judge Agreement Matrix (Mode: [inter_judge](#))

Setup: Pairwise Cohen's κ dan Agreement % untuk semua $C(11,2) = 55$ pasang judge.

Hasil κ antar judge (terpilih):

Judge A	Judge B	κ	Agree %	Both Pos %	Both Neg %
ChatGPT	DeepSeek-V3	0.7069	88.6%	20.7%	67.9%
ChatGPT	Qwen2.5-7B	0.6823	87.1%	—	—
DeepSeek-V3	Qwen2.5-7B	0.6xxx	~86%	—	—
DeepSeek-V3	Human	0.4219	75.7%	17.8%	57.9%
ChatGPT	Human	0.3777	74.4%	15.8%	58.6%
Qwen2.5-7B	Human	0.3767	73.2%	17.9%	55.2%
Gemma2-vllm-zs-bing	Human	0.3778	70.9%	21.9%	49.0%
Llama3-default	Human	0.2103	55.8%	27.2%	28.6%

Positive rate per judge:

Judge	Pos Rate	n_pairs
Human	1.000*	3,088
Llama3-default	0.6666	9,668
Llama3-vllm-zs-basic	0.5806	9,668
Gemma2-vllm-fs-basic	0.5357	9,668
Gemma2-vllm-zs-bing-strict	0.4529	9,668
Llama3-vllm-fs-basic	0.4489	9,668
Gemma2-vllm-zs-basic	0.4536	9,668
Gemma2-vllm-zs-bing	0.4103	9,668
Llama3-strict	0.3879	9,668
Llama3-vllm-fs-bing	0.3490	9,668
Qwen2.5-7B	0.3074	9,668
DeepSeek-V3	0.2799	9,668
ChatGPT	0.2494	8,750

*Human pos rate = 1.0 karena format positive-only; 3,088 pairs semua score=1

File: [results/final/extended/inter_judge_kappa.csv](#), [inter_judge_kappa_heatmap.png](#), [inter_judge_agree_heatmap.png](#), [inter_judge_posrate.csv](#)

5.2 Label Distribution per Rank Bin (Mode: [label_dist](#))

Setup: Untuk setiap kombinasi retriever × judge, hitung precision di setiap bin rank: [1-5], [6-10], [11-25], [26-50], [51-100].

Human precision (positivity rate) per retriever × rank bin:

Retriever	Rank 1-5	Rank 6-10	Rank 11-25	Rank 26-50	Rank 51-100
BGE-M3	0.280	0.093	0.034	0.012	0.004
Qwen3-embed	0.250	0.071	0.032	0.013	0.005
Hybrid BM25+BGE	0.215	0.079	—	—	—
Hybrid BM25+Qwen3	—	—	—	—	—
BM25	0.144	0.075	0.034	0.016	0.007
Qwen-embed (legacy)	0.003	0.001	0.001	0.002	0.001

Human precision = fraksi dokumen dalam rank bin yang muncul di human positive qrels

Interpretasi: BGE-M3 menempatkan 28% dari 5 dokumen teratas yang relevan menurut human (vs BM25 hanya 14.4%). Semua retriever menunjukkan gradien menurun yang jelas dari rank 1-5 ke rank 51-100, mengkonfirmasi kualitas ranking.

File: [results/final/extended/label_dist.csv](#) (330 baris: 6 retriever × 11 judge × 5 bin), [label_dist.png](#)

5.3 Full Retriever × Judge Matrix (Mode: `full_matrix`)

Setup: nDCG@10 untuk semua 66 kombinasi (6 retriever × 11 judge).

Lihat §4.3 untuk tabel lengkap. Summary tambahan dari `full_matrix.csv`:

Retriever Hybrid-BGE (BM25 + BGE-M3, nDCG@10 per judge):

Judge	nDCG@10
Human	0.5175
DeepSeek-V3	0.5233
Qwen2.5-7B	0.5062
ChatGPT	0.4814

Hybrid-BGE = 0.5191 dari evaluasi langsung, sedikit berbeda karena perbedaan versi candidates file

File: `results/final/extended/full_matrix.csv`, `full_matrix_ndcg_heatmap.png`,
`full_matrix_map_heatmap.png`

5.4 Error Analysis (Mode: `error_analysis`)

A. Query Difficulty Distribution

Category	Kriteria	Count	%
Hard	$\max \text{nDCG@10} < 0.1$	73	7.6%
Medium	$0.1 \leq \max \text{nDCG@10} < 0.3$	67	7.0%
Easy	$\max \text{nDCG@10} \geq 0.3$	820	85.4%

73 query (7.6%) gagal total di semua sistem ($\text{nDCG@10} = 0$ untuk semua retriever). Ini kemungkinan karena:

- Dokumen relevan tidak ada di Wikipedia Indonesia
- Query memerlukan reasoning multi-step
- Pooling bias (dokumen relevan tidak pernah di-retrieve oleh sistem yang membentuk pool)

File: `results/final/extended/hard_queries.csv`

B. Judge Disagreement per Query

qid	mean_disagreement	n_docs	frac_docs_disagreed
3594	0.2248	10	1.00
4865	0.2179	10	1.00
1091	0.2176	9	1.00
5009	0.2166	10	1.00

Query-query ini memiliki disagreement tertinggi di antara semua judge — semua 10 dokumen dinilai berbeda oleh judge yang berbeda. Ini adalah kasus borderline relevance yang genuinely sulit.

File: `results/final/extended/judge_disagreement.csv`

C. Reranker Failure Analysis

System: Qwen3-embed + BGE-hardneg-rk vs Qwen3-embed baseline (human qrels)

Category	Kriteria	Count	%
Big failure	$\delta < -0.2$	228	23.8%
Neutral	$-0.05 \leq \delta \leq 0.05$	291	30.3%
Big gain	$\delta > 0.2$	161	16.8%

Worst failures ($\delta = -1.0$): Query 1770, 2329, 3738, 3901, 4123, 4180, 4699, 4979, 5629 — baseline nDCG=1.0 → reranked nDCG=0.0. Ini terjadi karena query yang hanya memiliki 1 dokumen relevan, dan reranker mendorong dokumen tersebut keluar dari top-10.

File: `results/final/extended/reranker_failure.csv`, `error_analysis.md`

6. Kontribusi Per Anggota

Faiz (E-series)

Eksperimen	Detail	Output
Qwen2.5-7B Judge	Full inference pipeline, 51k pairs (train+val+test)	<code>qrels/qwen_*.txt</code>
Prompt ablation	5 varian prompt × Qwen2.5-7B	<code>kappa_prompt_ablation.csv</code>
Calibration	Threshold sweep $\tau=1,2,3$	<code>calibration_qwen.csv</code>
LoRA SFT	Fine-tuning Qwen2.5-7B sebagai judge (TRL SFTTrainer, Unsloth)	<code>kappa_qwen_lora_test.csv</code>
ORPO prep	Script + data preparation (belum ada hasil final)	<code>lora/train_orpo.py</code>
BM25 + Qwen-rk eval	CrossEncoder reranker dari Qwen qrels	<code>bm25_qwen_rk.json</code>
Extended analysis	Script <code>evaluation/extended_analysis.py</code> , semua 4 mode	<code>results/final/extended/</code>
WRAP_UP.md	Dokumentasi komprehensif assets	<code>results/WRAP_UP.md</code>

Arvin

Eksperimen	Detail	Output
BM25 retrieval	bm25s, no Java dependency	<code>candidates/bm25_*_top100.jsonl</code>
BGE-M3 dense retrieval	BAAI/bge-m3 + FAISS, semua split	<code>candidates/bgem3_*_top100.jsonl</code>
Hybrid BM25+BGE-M3	RRF fusion	<code>candidates/hybrid_test_top100.jsonl</code>
DeepSeek-V3 judge	API inference, 9,668 pairs	<code>results/qrels/deepseek_test.txt</code>
ChatGPT judge	gpt-4o-mini API, 8,750 pairs	<code>results/qrels/chatgpt_test.txt</code>

Eksperimen	Detail	Output
LoRA SFT (TRL)	Refactor ke TRL SFTTrainer, setup_and_train.sh	lora/

Karolina (Karol)

Eksperimen	Detail	Output
Qwen3-embed retrieval	Qwen/Qwen3-Embedding-4B, no instruction, semua split	candidates/qwen3*_top100.jsonl
Qwen3-embed with instruction	val split ablation +0.059 nDCG@10	candidates/qwen3_instruct_val_top100.jsonl
Hybrid BM25+Qwen3	RRF fusion + eval	candidates/hybrid_bm25_qwen3_test_top100.jsonl
Hard negative mining (Qwen3)	53,727 triplets dari Qwen3-embed false positives	results/reranker_data/qwen/
Hard negative mining (BGE-M3)	130,961 triplets dari BGE-M3 false positives	results/reranker_data/bgem3_hardneg/
Reranker training (Qwen3 HN)	BGE-reranker-v2-m3 + Qwen3 hard negs	results/models/reranker_qwen/
Reranker training (BGE-M3 HN)	BGE-reranker-v2-m3 + BGE-M3 hard negs	results/models/reranker_bgem3_hardneg/
All reranker evals	12 kombinasi first-stage x reranker	Semua JSON di results/final/
RQ3 bias analysis	Full analysis, visualisasi, laporan	results/final/rq3_bias_analysis.md
Bias analysis scripts	evaluation/bias_analysis.py	3 mode: aggregate/perquery/overlap

Vincent (Radit)

Eksperimen	Detail	Output
SahabatAI-Gemma2 judge	Gemma2-9B inference vLLM, train+val+test (9,668+33,076+8,282)	results/qrels/sahabat_llama_*.txt
SahabatAI-Llama3 judge	Llama3-8B inference (default + strict prompt)	results/qrels/sahabat_llama_*.txt , results/qrels_strict/

Eksperimen	Detail	Output
Gemma2 prompt ablation	5 varian prompt × Gemma2 (vLLM)	kappa_prompt_ablation_full.csv
Llama3 prompt ablation	3 varian prompt × Llama3 (vLLM)	kappa_llama_vllm_*.csv
Size ablation (RQ2)	BGE reranker dilatih dari Gemma2 qrels, N=100,300,500,1000,full	ablation_summary.csv , size_*.json
Learning curve analysis	Kurva learning N vs nDCG@10 vs Val AP	learning_curve.png , ap_vs_ndcg_curve.png
Reranker models × 5	Semua 5 model dari size ablation	results/models/reranker_{100,300,500,1000,full}/

7. Lokasi File & Artifacts

7.1 Qrels Files

File	Judge	n_pairs	Split
data/miracl-id/qrels/human/test.txt	Human	3,088	test
data/miracl-id/results/qrels/qwen_test.txt	Qwen2.5-7B	9,668	test
data/miracl-id/results/qrels/qwen_train.txt	Qwen2.5-7B	33,076	train
data/miracl-id/results/qrels/qwen_val.txt	Qwen2.5-7B	8,282	val
results/qrels/deepseek_test.txt	DeepSeek-V3	9,668	test
results/qrels/chatgpt_test.txt	ChatGPT (gpt-4o-mini)	8,750	test
results/qrels/sahabat_llama_test.txt	SahabatAI-Gemma2 (default)	9,668	test
results/qrels/sahabat_llama_train.txt	SahabatAI-Gemma2	33,076	train
results/qrels/sahabat_llama_val.txt	SahabatAI-Gemma2	8,282	val
results/qrels/sahabat-gemma_vllm_zeroshot_bing_test.txt	Gemma2 (vllm, zs-bing)	9,668	test
results/qrels/sahabat-gemma_vllm_zeroshot_bing_strict_test.txt	Gemma2 (vllm, strict)	9,668	test
results/qrels/sahabat-gemma_zeroshot_basic_test.txt	Gemma2 (partial)	2,151	test
results/qrels/sahabat-llama_vllm_fewshot_basic_test.txt	Llama3 (vllm, fs-basic)	9,668	test

File	Judge	n_pairs	Split
results/qrels/sahabat-llama_vllm_fewshot_bing_test.txt	Llama3 (vllm, fs-bing)	9,668	test
results/qrels/sahabat-llama_vllm_zeroshot_basic_test.txt	Llama3 (vllm, zs-basic)	9,668	test
results/qrels_strict/sahabat_llama_strict_test.txt	Llama3 (strict)	9,668	test
results/qrels_strict/sahabat_llama_strict_train.txt	Llama3 (strict)	33,076	train
results/qrels_strict/sahabat_llama_strict_val.txt	Llama3 (strict)	8,282	val

7.2 Candidates Files

File	Retriever	Split
candidates/bm25_test_top100.jsonl	BM25	test
candidates/bm25_train_top100.jsonl	BM25	train
candidates/bm25_val_top100.jsonl	BM25	val
candidates/bgem3_test_top100.jsonl	BGE-M3	test
candidates/bgem3_train_top100.jsonl	BGE-M3	train
candidates/bgem3_val_top100.jsonl	BGE-M3	val
candidates/hybrid_test_top100.jsonl	Hybrid BM25+BGE-M3	test
candidates/qwen_test_top100.jsonl	Qwen-embed (legacy)	test
candidates/qwen_train_top100.jsonl	Qwen-embed (legacy)	train
candidates/qwen_val_top100.jsonl	Qwen-embed (legacy)	val
candidates/qwen3_test_top100.jsonl	Qwen3-embed (no instr)	test
candidates/qwen3_train_top100.jsonl	Qwen3-embed	train
candidates/qwen3_val_top100.jsonl	Qwen3-embed	val
candidates/qwen3_instruct_val_top100.jsonl	Qwen3-embed (with instr)	val
candidates/hybrid_bm25_qwen3_test_top100.jsonl	Hybrid BM25+Qwen3	test

7.3 Model Artifacts

Model	Lokasi	HuggingFace	Training
BGE-reranker (Gemma2, N=100)	results/models/reranker_100/	—	3,874 triplets, 3 epoch
BGE-reranker (Gemma2, N=300)	results/models/reranker_300/	—	10,570 triplets

Model	Lokasi	HuggingFace	Training
BGE-reranker (Gemma2, N=500)	results/models/reranker_500/	—	18,346 triplets
BGE-reranker (Gemma2, N=1000)	results/models/reranker_1000/	—	37,018 triplets
BGE-reranker (Gemma2, full)	results/models/reranker_full/	—	121,500 triplets
BGE-reranker (Qwen HN)	results/models/reranker_qwen/	fassabilf/umbrela-ir-reranker-qwen	53,727 triplets (Qwen HN)
BGE-reranker (Qwen smoke)	results/models/reranker_qwen_smoke/	—	Smoke test only
BGE-reranker (BGE-M3 HN)	results/models/reranker_bgem3_hardneg/	karolinajocelyn/umbrela-ir-models	130,961 triplets (BGE HN)
LoRA Qwen2.5-7B (smoke)	results/models/lora_qwen_smoke/	fassabilf/lora-qwen-miracl-id-smoke (📦)	33k pairs SFT

7.4 Script Utama

Script	Fungsi
evaluation/eval_pipeline.py	Evaluasi first-stage + reranking (nDCG@10, MAP@10, R@100)
evaluation/eval_retrieval.py	Evaluasi first-stage saja
evaluation/metrics.py	parse_qrels() , compute_kappa()
evaluation/bias_analysis.py	Bias analysis (3 mode: aggregate/perquery/overlap)
evaluation/calibrate.py	Threshold sweep untuk binarisasi
evaluation/logit_inference.py	Logit-based scoring (optional)
evaluation/extended_analysis.py	Extended analysis (4+1 mode)
evaluation/plot_ablation_analysis.py	Plot size ablation
evaluation/plot_learning_curve.py	Plot learning curve
qrel_generation/inference.py	Inference LLM judge (standard)

Script	Fungsi
qrel_generation/inference_vllm.py	Inference vLLM (support --lora-path)
qrel_generation/judges.py	Registry judge models
reranker/prepare_data.py	Persiapan training data reranker (dari LLM qrels)
reranker/train.py	Training CrossEncoder reranker
reranker/inference.py	Inference reranker → TREC run
reranker/mine_hard_negatives.py	Hard negative mining dari candidates
lora/prepare_data.py	SFT data prep (human qrels → prompt/response)
lora/train.py	LoRA SFT training (manual loop)
lora/prepare_orpo_data.py	ORPO data prep
lora/train_orpo.py	ORPO training (TRL ORPOTrainer)

7.5 Extended Analysis Output

File	Konten
results/final/extended/inter_judge_kappa.csv	55 pasang κ + agree% + breakdown (56 baris)
results/final/extended/inter_judge_kappa_heatmap.png	11×11 heatmap Cohen's κ
results/final/extended/inter_judge_agree_heatmap.png	11×11 heatmap Agreement %
results/final/extended/inter_judge_posrate.csv	Positive rate per judge
results/final/extended/label_dist.csv	330 baris (6 retriever × 11 judge × 5 bin)
results/final/extended/label_dist.png	Grouped bar chart label distribution
results/final/extended/full_matrix.csv	66 kombinasi nDCG@10/MAP@10/R@100
results/final/extended/full_matrix_ndcg_heatmap.png	Heatmap nDCG@10
results/final/extended/full_matrix_map_heatmap.png	Heatmap MAP@10
results/final/extended/hard_queries.csv	960 queries dengan kategori hard/medium/easy
results/final/extended/judge_disagreement.csv	Per-query judge disagreement
results/final/extended/reranker_failure.csv	Query yang nDCG turun setelah reranking
results/final/extended/error_analysis.md	Ringkasan error analysis

7.6 HuggingFace Assets

fassabilf (Faiz):

Asset	Tipe	Status
fassabilf/umbrela-indo-ir	Dataset (results, qrels)	 Public
fassabilf/umbrela-indo-ir-results	Dataset (results detail)	 Private

Asset	Type	Status
fassabilf/umbrela-indo-ir-reranker-qwen	Model (BGE-reranker Qwen HN)	 Public
fassabilf/qwen-reranker-miracl-id	Model (BGE-reranker, MIRACL-ID)	 Private
fassabilf/lora-qwen-miracl-id-smoke	Model (LoRA Qwen2.5-7B)	 Private
fassabilf/orpo-qwen-miracl-id-smoke	Model (ORPO LoRA)	 Private
fassabilf/orpo-qwen-miracl-id-smoke-b3	Model (ORPO LoRA variant)	 Private

karolinajocelyn (Karol):

Asset	Type	Status
karolinajocelyn/umbrela-indo-ir-models	Model (BGE-hardneg-rk, Gemma2-rk)	 Public
karolinajocelyn/umbrela-indo-ir-data	Dataset (hard negative triplets)	 Public

arya-raditya (Radit):

Asset	Type	Status
arya-raditya/bge-reranker-gemma2-n100	Model (Gemma2-rk, N=100)	 Public

Catatan Reproduksi

Urutan Eksperimen

1. Setup data
 - └─ data/miracl-id/ (download dari MIRACL)
2. First-stage retrieval
 - └─ python retriever/run_bm25.py → candidates/bm25_*
 - └─ python retriever/run_bgem3.py → candidates/bgem3_*
 - └─ python retriever/run_qwen3.py → candidates/qwen3_*
3. LLM Judge inference
 - └─ python qrel_generation/inference_vllm.py --judge qwen → qrels/qwen_*
 - └─ python qrel_generation/inference_vllm.py --judge gemma2 → qrels/sahabat_llama_*
 - └─ [API] deepseek, chatgpt → qrels/deepseek_*, chatgpt_*
4. Kappa evaluation
 - └─ python evaluation/metrics.py → kappa_*.csv
5. Reranker training
 - └─ python reranker/prepare_data.py (dari LLM qrels)
 - └─ python reranker/mine_hard_negatives.py (dari candidates)
 - └─ python reranker/train.py → models/reranker_*/
6. Retrieval evaluation
 - └─ python evaluation/eval_pipeline.py → results/final/*.json
7. Bias analysis
 - └─ python evaluation/bias_analysis.py --mode all → results/final/bias_analysis/

8. Extended analysis

```
└─ python evaluation/extended_analysis.py --mode all → results/final/extended/
```

Dependency Utama

```
pip install sentence-transformers ranx bm25s faiss-cpu \  
vllm scikit-learn matplotlib seaborn \  
huggingface-hub trl peft unsloth
```

Atau untuk environment yang managed (macOS dengan Homebrew Python):

```
uv run --with ranx --with scikit-learn --with matplotlib --with seaborn \  
python3 evaluation/extended_analysis.py --mode all
```

Dokumen ini digenerate otomatis dari data eksperimen, diperbarui terakhir: 2026-05-28